

waveSense: Low Power Voxel-tracking Technique for Resource Limited Devices

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Figure 1: a) *waveSense*, a system to enable hand-gesture interactions on devices with limited power and processing capability. b, c) Proof of concept applications that extend the interaction space of a Virtual Reality (VR) headset. d) Samsung Gear VR headset modified with *waveSense*.

ABSTRACT

In this paper, we describe *waveSense*, a power-efficient optical voxel tracking system for hand tracking in resource-limited devices. *waveSense* uses off-the-shelf, low-cost, non-focused infrared (IR) sensors and emitters. To achieve power efficiency, we introduce a novel selective volumetric illumination (SVI) technique, which illuminates different areas of a scene with variable power IR light and captures the reflected energy using non-focused IR sensors. Our technical evaluation suggests the feasibility of using *waveSense* for hand tracking performed within 45cm radius of the device. We believe with *waveSense*, gesture-based interactions would become more practical on resource-constrained wearables such as head-mounted displays (HMDs) and smartwatches.

CCS CONCEPTS

• Human-centered computing → Interaction techniques; Gestural input.

KEYWORDS

Gesture Recognition, Interacting with small devices, Selective Volumetric Illumination, Smart Wearables

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1 INTRODUCTION

The introduction of virtual reality (VR) devices, such as *Samsung Gear VR*, require interaction methods that go beyond traditional pointing technologies, such as buttons and touch-screens [8, 17, 25]. Being immersive and entertaining, gestural interactions provide an ideal input modality in the context of VR. Although real-time 3D motion tracking of hands is an essential component in gestural interactions, these systems often require dedicated computer vision systems with complex computation power needs [22].

Our main goal with *waveSense* is to develop a low power and sparse hand tracking system with low processing overhead for gesture recognition for resource-limited wearable VR glasses, such as *Samsung Gear VR*. While there are other low-power gesture sensing systems available, they are either limited in the range [29, 30] or needs high processing power [6]. As a solution, we propose a trade-off between hand tracking resolution to save power while being able to recognize a reasonable number of distinct points in space for gestural interactions. To this end, in *waveSense*, we introduce *selective volumetric illumination (SVI)*, a novel illumination technique that uses variable emission powers and digital (binary) sensors to achieve low power hand tracking with small data-throughput. This results in an extended sensing range, improved immunity to background noise, and lower energy consumption compared to other alternatives [30].

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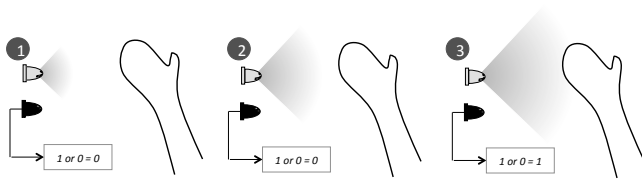


Figure 2: Working principle of the SVI technique. From a-1 to a-3, a scene with a target (hand) is illuminated by a variable power IR emitter, creating a volumetric illumination pattern. Once the illumination region reaches the target (a-3), the sensor produces a digital 1 output.

In our *Selective Volumetric Illumination* scheme, we use off-the-shelf IR sensors to emit modulated pulses of IR energy at different illumination intensities. As shown in Figure 2, these different intensities illuminate different spacial regions in space. The presence of any IR opaque targets (e.g., a hand) would reflect a portion of this energy back to the sensors. Since the emitted power is modulated with a known frequency, at the sensor, this reflected energy can be efficiently extracted against background noise to identify the presence of a target as a digital (binary) 1 or 0. Since the emitted intensity is modulated, each binary value represents a unique combination of the travel distance of light from the transmitter to the sensor. Consequently, a spatially distributed combination of multiple sensors and emitters under the scheme can be used to discriminate the target voxel in the sensing range efficiently.

We use the SVI technique to implement a hand-gesture recognition proof of concept prototype on a Samsung Gear VR headset. This proof of concept system was able to achieve a comparable hand tracking accuracy and higher power efficiency compared to *zSense* while extending the sensing range to $\sim 45\text{cm}$. The contributions of this paper are:

- **Introduction of a Selective Volumetric Illumination (SVI) for power-efficient gesture recognition.** The proposed SVI technique achieves high power efficiency by adaptive illumination of emitters from low to high intensities.
- **Adaptation of a digital sensing technique with a modulated IR signal.** This provides two key advantages: 1) a longer sensing range while preserving power, and 2) a low data throughput requiring less processing power for tracking targets.
- **Prototype implementation of *waveSense*.** This includes the technical details of the implementation, verification of its performance through a systematic technical evaluation and exploration various applications using a virtual reality headset.

2 RELATED WORK

2.1 Camera-based Gesture Recognition

Regular 2D cameras [8, 20], markers [3, 24], and commercial depth cameras¹ have frequently been used in literature for their ability

to track gestures robustly in real time [2]. For example, in Omni-Touch [10], Harrison et al. used a depth camera attached to the shoulder to identify various gestures on surfaces for interaction in the air. In [15], Jones et al. use a similar depth camera tracking system to provide around the device interaction to investigate free-space interactions for multi-scale navigation with mobile devices. Song et al. [26] introduced a gesture technology for static hand gestures as opposed to dynamic gestures in *waveSense*. Typically, such computer vision-based approaches require high computational processing power and high energy, which makes them less desirable for resource-constrained devices.

2.2 Magnetic or Electric Field Based Gestures

Magnetic field-based sensing is another frequently used technique to extend the interaction space around mobile devices [1, 5, 16]. Works, such as [11, 14], use external permanent magnets to extend the interaction space around the mobile device. Both works use the inbuilt magnetometers of the device to detect the magnetic field changes around the device as input to the system. In [11], authors use a magnet on the finger for gestural input, and [14] introduces a new un-powered device for interaction. However, the magnetic sensing approach requires instrumenting the user – generally having the user to wear a magnet on the fingertip.

2.3 Sonic Wave Based Sensing

TapSense [12] uses an embedded sound sensor on mobile devices that allow users to perform different interactions by tapping with various parts of the finger, such as nail, knuckle, tip, etc. Skin-put [13], on the other hand, uses the human body for acoustic transmission, where authors introduce a sensor embedded in an armband to identify and localize vibrations caused by taps on the body as a form of input. In both scenarios, physical contact between the device and the sensor is necessary to recognize the input. In *waveSense*, our focus is to extend the interaction area beyond the surface.

Doppler effect based gesture sensing was introduced in *SoundWave* by Gupta et al. [7]. While *SoundWave* claims it can be adapted to mobile settings, the number of gestures it can identify remains limited because the Doppler effect is only functional radially from the sensor. Also, commercial apps like *wave-o-rama* only work with few basic gestures, such as swipes, while *waveSense* is capable of tracking complex, expressive gestures. Off-the-shelf commercial capacitive sensing ICs (such as Microchip) have a much lower sensing range ($\sim 10\text{cm}$) and need large antennas. These are not desirable for gesture-based interactions on head-mounted displays where the range typically exceeds 20cm .

2.4 IR, Radio waves and non-focused sensing

Infrared-based approaches have been used to sense touch and to extend the interaction space of mobile devices [4, 18, 19, 23, 27, 28].

In SideSight, the authors use arrays of infrared sensors attached on the two sides of a mobile device to provide multi-“touch” interaction when placed on a flat surface [4]. Similarly, in [23], the authors use infrared beams reflected from the back of your hand to extend interactions with a smart wristwatch. In Gesture Watch [18], the authors use infrared proximity sensors on a wrist-worn device

¹<https://www.ultraleap.com/tracking/>

combined with hidden Markov models to recognize gestures for interaction with other devices. Taylor et al. use an array of 64 IR transceivers to detect gestures in the range of 33mm above a keyboard [28]. Koyama et al. proposed another IR array-based solution for in-car multi-touch and gesture sensing[19].

However, since these projects use linear sampling with IR sensors, a high density of emitters and sensors is necessary to track gestures from a relatively small space making them power inefficient. Soli is another high performing gesture sensor developed for mobile applications [29]. Soli enables the sensing of micro gestures within a short-range from the sensor, while *waveSense* is designed for macro hand gestures. In contrast, Mime [6] introduces a 3D gesture sensor that uses a three-pixel infrared time-of-flight module combined with an RGB camera. However, measuring time-of-flight requires extremely high sampling frequencies, data conditioning, and processing, which is usually not available in wearable devices. Non-linear spatial sampling (NSS) for gesture recognition was first introduced by *zSense* [30], where a shallow depth gesture recognition system was developed using a comparatively small number of sensors and emitters to recognize finger gestures. However, *zSense* is based on analog energy reflection measurements, which is highly vulnerable to noise, and the sensing range is relatively much smaller ($\sim 15cm$). In *waveSense*, we managed to overcome these limitations by selective illumination of volumetric regions (SVI) using a digital sensing method.

Additionally, there are some commercially available sensors that are designed for off-the-shelf gesture sensing. Some notable examples include Microchip GestIC, Avago APDS9960, and AMS TMG4903. These ICs are power-optimized, miniature, and designed for sensing basic gestures such as swipes in short-ranges ($\sim 10 - 30cm$). The main advantage of *waveSense* compared to these technologies is that since our system can track targets (voxels), gestures are not pre-defined, and the designer can choose customized gestures according to the application requirements. Furthermore, *waveSense* can be used for sparse pointing tasks.

3 WAVESENSE CONCEPT

The fundamental idea of SVI is to illuminate selected volumetric regions of the interested scene and collect a non-linear sample of the reflected energy from targets. SVI uses IR emission power as the primary mode of controlling the illumination and the changes in the direction of sensors, emitters, or both as the secondary mode of control. Compared to traditional camera-based approaches or linear sampling approaches [4, 18], SVI needs a much lower number of sensors (generally two to six).

In prior works, such as *zSense* [30], IR sensors are switched at a constant power level to create a spatial pattern. Reflected light intensity from the target is measured as an analog value to estimate the target location, which works efficiently for a shallow depth system. However, to increase the sensing range, it will need to employ high power LEDs (defeats the purpose of power saving) and risk being affected by external noise.

In contrast, the SVI technique systematically illuminates different volumetric areas of a scene with modulated infrared (IR) light with variable power levels. The digital sensor records the reflected light as a binary bitstream. One frame of data-capture includes all the

possible combinations of power levels $\times$ emitters. For example, in the configuration shown in Figure 2, after the three power levels, the system will record a data frame with 3-bits, [0, 0, 1]. This approach allows for an increased sensing range.

Furthermore, since the emitters are modulated with a pre-defined frequency, sensing is highly immune to background noise. Therefore, emission requires much less illumination power compared to the analog approach [30]. This collective modulation of the sensing environment with illumination results in each sensor identifying a target as a binary value indicating its presence (See Figure 2). Therefore, only a small dataset is needed to track the location of a target voxel, further reducing the power and processing requirements.

In *waveSense*, relative spatial displacements (linear or angular) between physical locations and directions of sensors and emitters play an important role in determining the volumetric illumination pattern. Changing the physical location and temporal emitter power levels can create complex illumination patterns in a sensing scene. Figure 3 shows three spatial configurations of a sensor and an emitter. Each emitter operates at four (4) power levels. Sub-figures 3a, 3b and 3c show cross-sections of the illumination patterns for linearly displaced sensor locations along one axis.

As shown in the Figure 3, the spatial configuration of sensors and emitters is a crucial factor in determining the modulated spatial illumination pattern and the quality of the captured signal. Therefore, a *waveSense* sensing configuration must have at least two IR emitters coupled with one or more sensors. The accuracy and the number of trackable voxels, as well as the recognizable gestures, change with the number of sensors and emitters. Depending on the interaction requirements and resources available, a trade-off is required between the number of desirable gestures and the number of IR sensors/emitters. In our current prototype, we have selected 2 emitters, 6 sensors, and the number of power levels as 8 (inclusive of the "OFF" state) based on iterative testing to fit the sensitive area. Our system resulted in a range of 45cm, which can be increased by using stronger emitters with higher illumination intensity. However, for VR application, the 45cm range is sufficient to track the users' own hand gestures.

4 IMPLEMENTATION

We implemented a proof of concept prototype of *waveSense* using self-contained IR receivers (VISHAY TSDP34356) and emitters (VISHAY TSAL6200). This sensor and emitter arrangement is designed in such a way to allow easy attachment onto wearable devices, such as Samsung Gear VR. The functional block diagram of *waveSense* is shown in Figure 4a. The system is controlled using Nordic's nRF51822 (16MHz, 3.3V), a System-on-a-Chip (Figure 4d) which has built-in Bluetooth Low Energy (BLE). The microprocessor is used to modulate the IR emission at 57.6kHz to increase the immunity to background noise. A power controlling circuit, combined with the modulated signal mentioned above, is used to achieve SVI by controlling the power supplied to the emitters at 8 power levels. Simultaneous operation of emitters is avoided to reduce synchronization issues in modulation, thus 4 levels $\times$ 2 emitters resulting 16 patterns (inclusive of both emitter "OFF" state to account for noise introduced by environment). For each pattern, each sensor records one bit with either digital 1 or 0. Our prototype

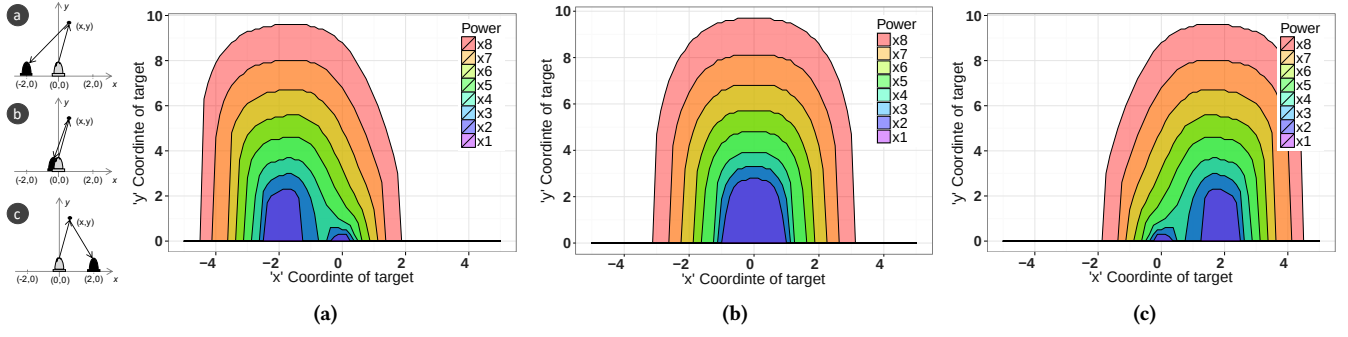


Figure 3: Cross-sections of regions of illuminations for eight different power levels. Corresponding emitter and sensor locations are shown in smaller sub-figures in the left.

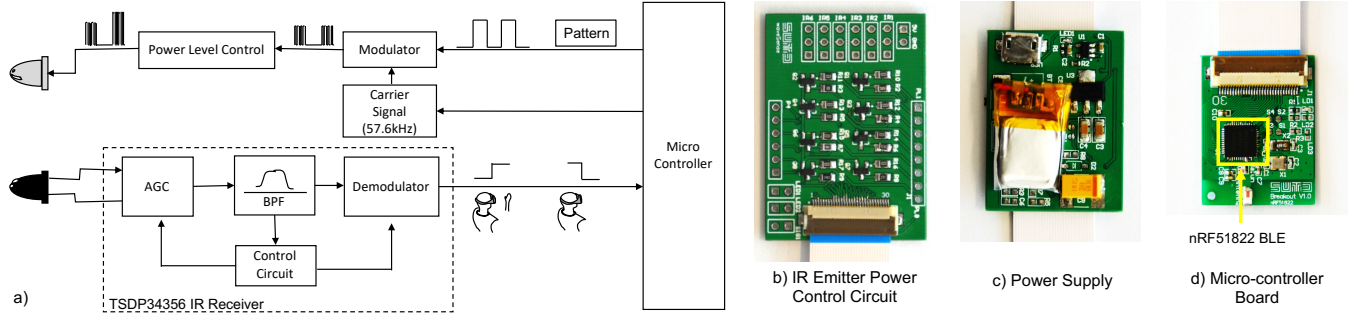


Figure 4: Block diagram of the hardware system

consists of 6 sensors, resulting in $16 \times 6 = 96$ bits of data per frame. At the operating frame rate of 40 fps , the system produces a data throughput of $96 \times 40 = 3840$ bits per second (or 480 Bps).

Each SVI pattern is kept on for $210 \mu\text{s}$, and the reflected light is captured by the IR sensors, which are self-contained with an *automatic gain controller (AGC)*, *demodulator* and the other required circuitry to detect the emitted light even in a noisy environment. The output of the IR sensors is a digital signal which can be directly fed into the microprocessor using its General Purpose Input Output (GPIO) pins. This self-contained sensor eliminates the need for an amplifier circuit, which is essential if a generic photodiode is used (such as in *zSense*) [30]. Furthermore, it enables the emitters to be operated at a low power levels spanning from 1.05 mW to 18.3 mW (average 9.7 mW). Total ON time per second for emitters is 0.067 s ($210 \mu\text{s}$ pattern duration $\times 16$ patterns $\times 40$ times per second $\times 0.5$ due to modulation). Thus, the total power consumption of emitters is 0.65 mW ($9.7 \text{ mW} \times 0.067 \text{ s}$). Maximum power consumed by the sensors is 8 mW (10 mW Max power $\times 0.13 \text{ s}$ on time $\times 6$ sensors). Therefore, the power consumption of the sensing system (excluding the microprocessor) is 8 mW . The microprocessor is in sleep mode 50% of the time, where its power consumption is 0.5 mW (according to the datasheet). As such, the total power consumption of the sensor module adds up to 9.15 mW . In our prototype, the gesture classification algorithm runs on either a computer or a mobile phone, making it difficult to measure the power consumption of that part accurately. However, as mentioned earlier, the low data rate of 480 Bps is indicative of the low processing power required.

4.1 Data Processing and Machine Learning

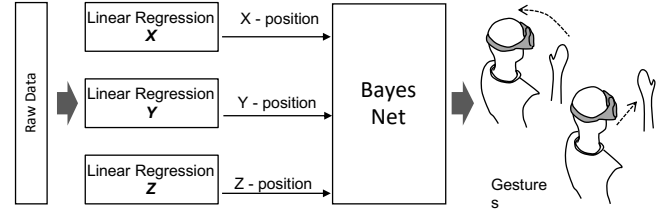


Figure 5: Data processing block diagram

For applications, the identification of gestures is done in two stages: 1) track the x, y, z locations of a target using raw data, 2) the x, y, z locations are sent to another classifier using a time-based moving window to recognize the gestures (Figure 5). The first step uses a linear regression model to identify the continuous x, y, z locations. The second step uses the BayesNet machine learning algorithm to identify the gestures using x, y, z locations.

In the first stage, received IR sensor data is captured as a 96 bit array. This array is pre-processed to minimize noise effects by removing any bit strings with only less than 5% of active bits. Then, the x, y, z locations are calculated using an independent regression model for each axis. These models are pre-trained databases to extract the location of the target using an artificial target hand. We use the SMO method to partition the training problem into

smaller problems that can be solved analytically using heuristics. The first stage of the training process only depends on the hardware prototype and does not require a per-user-training. The specific procedure of training the system is described in the evaluation section. When a user performs a gesture, 30 consecutive locations of the hand (from the previous classification) will be fed to the second stage of the classification module to recognize the gesture. This stage requires per-user-training. The classification systems are implemented with the BayesNet algorithm provided by the Weka Toolkit [9] and interfaced using a Java application. The software development was done using Android Studio, Unity 5, and Java API.

5 EVALUATION

We evaluated the feasibility of the volumetric illumination concept with respect to its accuracy in decoding location data of a reflected target. Training data was collected by moving an artificial hand in a 3D grid consisting of 126 points ($7 \times 9 \times 2$), on an imaginary coordinate system marked by unit lengths of $x = 5cm$, $y = 5cm$, $z = 20cm$ respectively. For each location, 50 data samples of sensor data were recorded. The sensor was placed at the location (4, 0, 0.5). The 3d region selected is sufficient to capture the movement range of hands.

5.1 Data Analysis and Results

Data is labeled to represent Cartesian points marked by respective x , y , and z axes values of the artificial hand. Axes x , y , and z respectively represent the axes along the frontal plane of the torso of the user (i.e., left to right), perpendicular to the frontal plane (i.e., depth) and vertical axes (i.e., up-down). Altogether, 12348 data samples were recorded corresponding to 126 locations and 50 samples each. Each sample consists of a 96bit data array. In our initial examination of the data set, we noticed that when the hand is moved along close parallels to the sensor ($y \leq 15cm$), either most of the bit values are 1 (all sensors saturated, $10cm < x < 20cm$) or 0 (hand is vertically out of range, $x < 10cm$ or $x > 20cm$). Therefore, we removed this range from the data set used for further analysis. However, this does not affect the range of the system since only the values close to the sensor are removed. This resulted in 98 data grid points for further analysis ($7 \times 9 \times 2$).

5.1.1 Data Modeling. As shown in Figure 3b, distance to the target, and the sensor response to reflected energy can be modelled in a polynomial function. Since each data bit in our data samples captures this polynomial characteristic, we hypothesize that a linear combination of bit values should be able to indicate the location of the target. In order to test this hypothesis, we model the data using three linear regression models to predict the x , y , and z values as continuous variables. We divided the grid points into two subsets, resulting in a training and a test dataset. The training dataset consist of combinations of coordinates where $x = 15, 25, 35, 45$, $y = 15, 25, 35, 45$ and $z = 0, 20$. Respectively, the test dataset consist of coordinates with $x = 20, 30, 40$, $y = 20, 30, 40$ and $z = 0, 20$. We use the `lm()` function from *R* toolkit (version 3.5.3) to model the data. The training dataset was used to generate the model, and the test dataset was used to evaluate the model.

5.1.2 Results. Regression results showed a significant correlation of data and the coordinates. Training regression models for x , y and z axes resulted in $R^2 = 0.96$ ($p < 0.001$), $R^2 = 0.70$ ($p < 0.001$) and $R^2 = 0.98$ ($p < 0.001$) respectively. Figures 6a, 6b and 6c shows the performance of the regression models for training data.

We used the developed model to predict the coordinates for the test dataset. Figures 6d, 6e and 6f respectively show the real and predicted coordinate values for the test dataset for each axes. Maximum average deviation of 2.84cm is observed at $x = 40cm$. While this error is not suitable for precise pointing tasks, since it is significantly smaller than the width of an average adult hand, we believe it is an acceptable error for pointing and gesturing with hands.

6 WAVESENSE APPLICATIONS

6.1 Image Gallery:

This application simulates the content browsing and selection task in a virtual reality device. We developed an image gallery application where users can browse with four distinct gestures (Figure 7-a). The images in the application are uniformly distributed around the user in a horizontal circle (Figure 7-b). When either *Close-Left-Swipe* or *Close-Right-Swipe* gesture is performed (figure 7-c and 7-d), either left image or right image of the user's view gains focus and moves to the center. To select the focused image in the front, the user can perform a *Middle-Pull* gesture, which moves the image closer to the user demonstrating the selection. *Middle-Push* gesture can be made in front of Gear VR for returning to the default browsing position.

6.2 First-Person Game:

This application emphasizes the potential of *waveSense* in immersive gameplay in a virtual reality context. For this, we developed a first-person game, where a user has to destroy incoming armorer-tanks with hand gestures. Tanks approach straight towards the user randomly from four directions with a speed that increases over time. The task given to the user is to destroy the tanks on four minefields marked in red (Figure 8-b) before they escape and attack the user. A user could accomplish this task using four different angled push gestures; *Left-Most-Push*, *Left-Push*, *Right-Push* and *Right-Most-Push* (Figure 8-a). If the user performs the respective gesture while the tank is on the respective minefield, they are destroyed. For demo purposes, a user wins the game when he or she destroys 20 tanks within 60 seconds, whereas the user loses when the number of escaped tanks accumulates to 10.

7 LIMITATIONS AND FUTURE WORK

The current system has a low spatial resolution, performs poorly at precise pointing, and is incapable of tracking multiple targets simultaneously. This is particularly prominent in the targets in the illumination boundaries between different power levels, and evident in Figures 6a to 6f. However, with the machine learning approach (as shown in [30]), our paper demonstrates that recognition of hand gestures is feasible with *waveSense*. Thus, we recommend the use of *waveSense* technology as a gesture recognition system rather than a high-fidelity tracking system. We envision a future hybrid system where *waveSense* is combined with high-resolution sensors (such

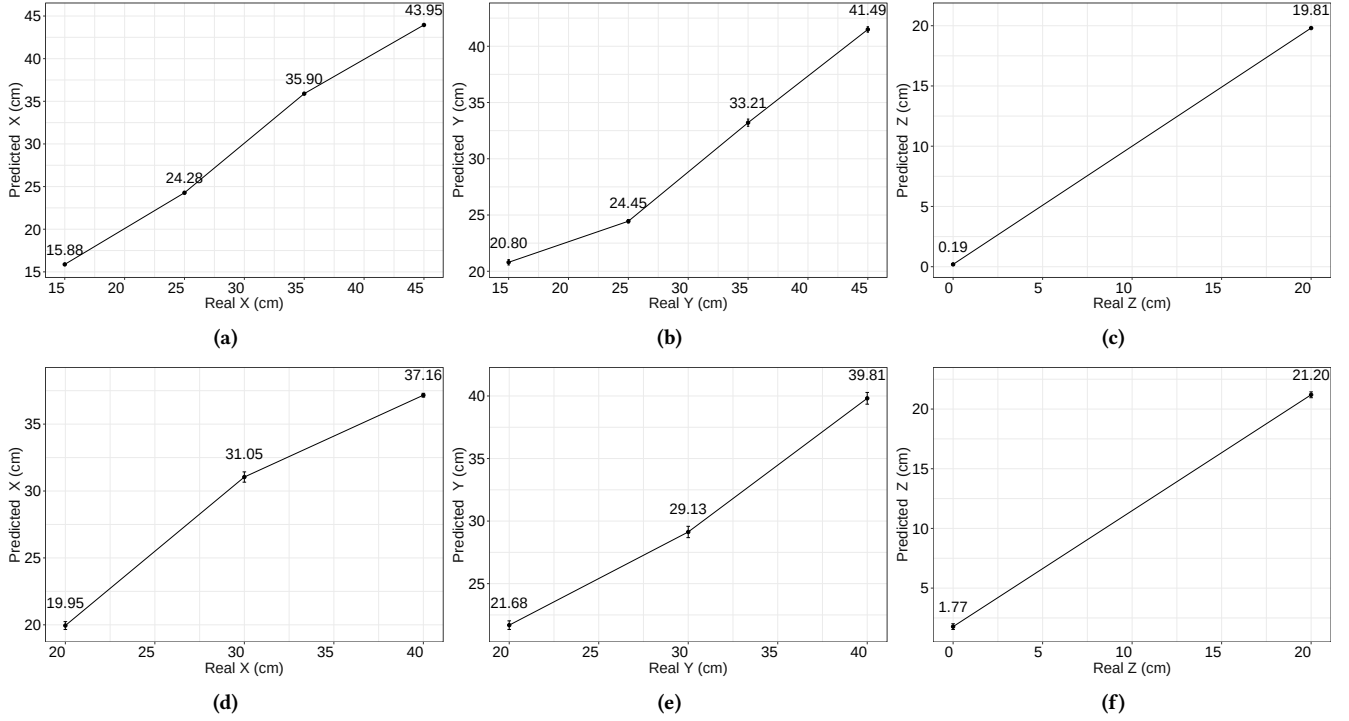


Figure 6: Results of the evaluation: a), b) and c) shows the predicted x , y and z locations for the training dataset using the regression model. c), d) and e) shows the predicted x , y and z locations for the test dataset using the regression model.



Figure 7: waveSense enabled Image Gallery Application. a) Gestures used. b) User view. c, d) Actual usage scenario.

as a depth camera) to enable the dynamic switching between high-resolution mode (with a depth camera) and power-saving mode (with *waveSense*), as shown in [6].

Although we have not conducted a user study on gestures, we have demonstrated the system at a conference where more than 100 people tried it with different skin colors (also different colors of shirt sleeves) without any noticeable effect.

We plan to do a formal evaluation of gesture recognition space in the future. This style of interaction can serve to enable reflexive interaction [21], as gestures are minor-complex and performed in a quick manner, thus easily internalized within motor memory.

In terms of system design, since the off-the-shelf IR emitter has a fixed carrier frequency of 57.6kHz, it is susceptible to other IR-operated home appliances, such as remote controllers. Future

development of the system would investigate the effect of carrier frequency on robustness against other IR light sources. It is also worthwhile to explore different modulation frequencies to increase the frame rate.

Another interesting direction is to explore the interaction between emitter/receiver arrangement and sensing capabilities (range and resolution), to find the sweet spot in the trade-off between resolution and resource consumption, particularly for mobile VR.

8 CONCLUSION

In this paper, we described *waveSense*, a power-efficient voxel tracking system for resource-limited devices. *waveSense* uses off-the-shelf, low-cost non-focused infrared (IR) sensors, and emitters. To

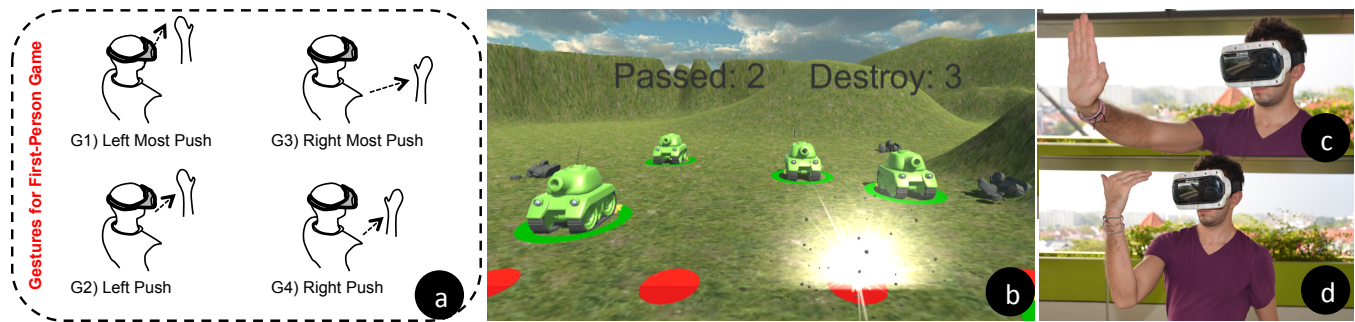


Figure 8: waveSense enabled First-Person Game a) Gestures used. b) User view. c, d) Actual usage scenario.

achieve power efficiency, a novel *Selective Volumetric Illumination* (SVI) technique was introduced, which selectively illuminates different areas of a scene with variable-power-IR-radiation and captures the activity using non-focused IR sensors. Our technical evaluation suggests the feasibility of using waveSense to track a hand within $\sim 45\text{cm}$ radius of the device using a simple linear regression approach with a maximum average deviation of 2.84cm from the actual target. We believe with waveSense, gesture-based interactions would become more practical on resource-constrained wearables, which would lead to better user experience on such devices.

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